

Challenge "Analyze an attack"

MD5sum: matched, the file integrity is intact.

```
(root@kali)-[/home/kali/Downloads]
# md5sum Attack_Logs.zip
63d96b5f2d742b159bffc7a512e29ff1 Attack_Logs.zip
```

Investigation Methodology

- The investigation was performed using Code OSS global search (Ctrl + Shift + F).
- Regex filters were used to search multiple keywords at once.
- Important indicators such as usernames, domains, processes, IP addresses, and suspicious executables were pivot points.
- Instead of reading entire logs, only attack-relevant timestamps and events were extracted.
- Attack stages were correlated chronologically to reconstruct the timeline.

Important Search Filters Used

- **POST|login|verify|Office365**
- **mimikatz|powershell|psexec|svc_admin**
- **steal|temp|\.exe**
- **delete|backup|shadow**
- **SUCCESS|LOGIN|VPN**

Question 1 – one phishing URL that was clicked by a user and resulted in credential theft. Document the full URL and timestamp based on your analysis.

The investigation identified a phishing campaign targeting Office365 users. Mail gateway logs showed that user l.moser@company.ch received a phishing email from no-reply@login-verification-alert.com with the subject "Security Notification: Unusual Login Attempt".

Proxy logs later showed that the victim accessed the phishing domain:

URL: http://login-verification-alert.com/session/check/Office365

Timestamp: 2025-07-14 11:48:37

Victim: l.moser@company.ch

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```
2025-07-14 11:48:37.000,l.moser@company.ch,GET,http://login-verification-alert.com/session/check/Office365,200,Mozilla/5.0  
2025-07-14 11:48:53.000,l.moser@company.ch,GET,http://update.mimikatz.org/download,200,Mozilla/5.0
```

Figure 1: Proxy log showing phishing URL access.

```
1:47:15,l.moser@company.ch,reply@login-verification-alert.com,"Security Notification: Unusual Login Attempt""
```

Figure 2: Mail gateway evidence of phishing email delivery.

```
{  
  "Timestamp": "2025-07-14 11:59:55",  
  "EventID": 4688,  
  "Level": "Information",  
  "Source": "Microsoft-Windows-Security-Auditing",  
  "Computer": "VENUS.corp.local",  
  "Account Name": "l.moser",  
  "New Process Name": "C:\\Users\\l.moser\\AppData\\Roaming\\mimikatz.exe",  
  "Command Line": "\"C:\\Users\\l.moser\\AppData\\Roaming\\mimikatz.exe\"",  
  "Message": "A new process has been created."  
},  
{  
  "Timestamp": "2025-07-14 12:02:10",  
  "EventID": 4624,  
  "Level": "Information",  
  "Source": "Microsoft-Windows-Security-Auditing",  
  "Computer": "VENUS.corp.local",  
  "Account Name": "svc_admin",  
  "Logon Type": "10 (RemoteInteractive)",  
  "Message": "An account was successfully logged on."  
},  
{  
  "Timestamp": "2025-07-14 12:07:50",  
  "EventID": 4688,  
  "Level": "Information",  
  "Source": "Microsoft-Windows-Security-Auditing",  
  "Computer": "VENUS.corp.local",  
  "Account Name": "svc_admin",  
  "New Process Name": "C:\\Windows\\Temp\\spread_payload.bat",  
  "Command Line": "\"C:\\Windows\\Temp\\spread_payload.bat\"",  
  "Message": "A new process has been created."  
}
```

Figure 3: Jump host logs showing Mimikatz execution and privilege escalation.

```
51 2025-07-14 11:49:12,l.moser@company.ch,85.203.22.19,10.1.57.128,OK,CISCO-VPN-ZH01
```

Figure 4: VPN log showing successful VPN access from attacker IP.

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```
"PID": 4378,  
"Name": "stealbit.exe",  
"User": "svc_backup",  
"StartTime": "2025-07-15 16:12:00",  
"Memory": "57426 K",  
"Path": "C:\\Users\\svc_backup\\AppData\\Local\\Temp\\stealbit.exe",  
"SHA256": "3E36B396C4CB71B8EAAE2300C21BEC26700B27CE5F68E83EF6B86D214E294C88"
```

Figure 5: Pluto process dump showing stealbit.exe exfiltration process.

Question 2 – Determine which IP address accessed the network using the stolen credentials. Document the IP address, the timestamp of the login, and the user account used.

User Account: l.moser@company.ch

Source IP: 85.203.22.19

Timestamp: 2025-07-14 11:49:12

Assigned Internal VPN IP: 10.1.57.128

VPN logs showed a successful remote VPN connection shortly after the phishing website visit, indicating the attacker used stolen credentials.

Question 3 – Determine which account was used to access the jump host and what specific action indicates privilege escalation. Document the username and describe the suspicious action with a timestamp.

Compromised User: l.moser

Privilege Escalation Tool: mimikatz.exe

Timestamp: 2025-07-14 11:59:55

Jump host logs showed execution of mimikatz.exe from the AppData\Roaming directory. Mimikatz is a well-known credential dumping tool used to extract administrator credentials and Kerberos tickets from memory.

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Shortly afterward, the privileged account svc_admin logged in remotely and executed spread_payload.bat from the Windows Temp directory, indicating lateral movement and malware deployment.

Question 4 – Identify the process responsible for data exfiltration from system Pluto. Document the process name, PID and briefly explain what makes it suspicious.

Process Name: stealbit.exe

PID: 4378

Start Time: 2025-07-15 16:12:00

The process stealbit.exe was executed from the Temp directory under the svc_backup account. The filename, execution path, and timing strongly indicate data exfiltration activity. StealBit is commonly associated with ransomware operators and exfiltration tooling.

Question 5 – Reconstruct the timeline of the attack using information from all provided logs. Document the attack sequence in exactly 5 steps with timestamps and brief explanations.

Timestamp	Attack Activity
2025-07-14 11:48:37	Victim l.moser visited phishing Office365 URL.
2025-07-14 11:49:12	Attacker accessed VPN using stolen credentials from IP 85.203.22.19.
2025-07-14 11:59:55	mimikatz.exe executed on VENUS.corp.local for credential dumping.
2025-07-14 12:02:10	svc_admin logged in remotely and spread_payload.bat was executed.
2025-07-15 16:12:00	stealbit.exe executed on Pluto for data exfiltration.

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Question 6 – Identify the step that is missing to complete a full double extortion attack. Briefly document the missing step.

The investigation revealed phishing, credential theft, privilege escalation, lateral movement, malware deployment, and data exfiltration. However, there was no evidence of ransomware encryption, ransom notes, or public extortion threats.

Missing Phase: Ransomware encryption / extortion phase